ACADEMIC PROJECT REPORT

Water Quality Prediction

Gradient Boosting and Deep Neural Networks

A chronological comparison on the source-defined surface-water index

Project 18 Team

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Repository: https://github.com/hypnoastic/Water_Quality_Prediction

Abstract and objectives

This project compares XGBoost and a regularized PyTorch multilayer perceptron for regression of the released CCME_Values index. The supplied five-country dataset contains 2,827,977 rows spanning 1940-2023. We remove 9,971 exact duplicates, apply declared input-quality rules and fit all preprocessing on a shared 300,000-row historical training sample. We separate 2019 validation, 2020 residual calibration and a complete 2021-2023 test partition of 202,886 rows.

XGBoost obtains test MAE 0.1099 and RMSE 0.4262. The selected 128/64-unit MLP obtains MAE 0.4466 and RMSE 0.9161. Validation selects a 90% XGBoost, 10% MLP blend, which obtains RMSE 0.4222 and R-squared 0.99861, with worse MAE than XGBoost. A source-formula audit reconstructs every released label within 1e-7. Consequently, the experiment establishes index emulation rather than independent environmental prediction.

Research question

How do a strong boosted-tree model and a small neural network compare when reproducing one documented continuous index under chronological evaluation?

Objectives

Define one target and trace its construction. Audit measurements, units and prior imputation. Compare both model families against mean and Ridge baselines. Explain model reliance and examine uncertainty across countries and index bands.

Keywords: water quality, XGBoost, MLP, regression, temporal evaluation, derived target, interpretability.

1. Dataset and exploratory analysis

Karim et al.'s Figshare v2 release [1] supplies eight chemistry measurements, four metadata columns and two related target columns. The combined CSV is the single training source. Country files are alternate views and must not be appended to it.

Country	Source rows	Distinct Area labels
England	2,129,198	48,642
USA	413,814	955
Ireland	235,019	9,737
China	45,997	1
Canada	3,949	2,354

England contributes 75.3% of rows. China has one Area label, so Area is an imperfect site identifier. Future test observations represent four countries, with only 46 Canadian rows and no Chinese rows.

Inputs: ammonia, biochemical oxygen demand, dissolved oxygen, orthophosphate, pH, temperature, nitrogen and nitrate. Concentrations use the supplied mg/L headers, temperature uses a Celsius header and pH is unitless. Turbidity, conductivity and dissolved solids are absent.

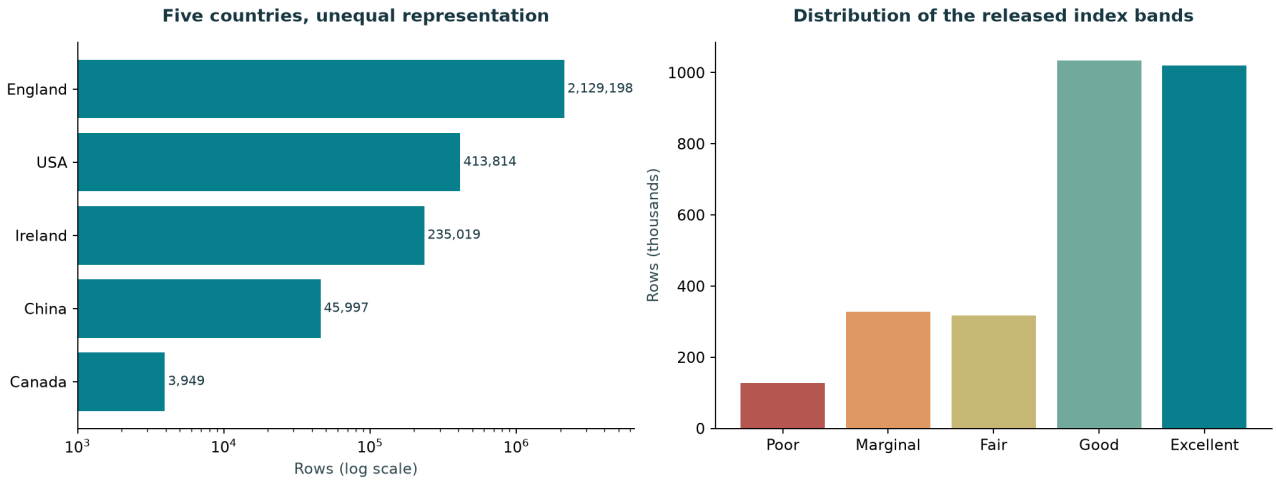


Figure 1. Country representation and released index bands. The country-count axis is logarithmic.

2. Target definition and formula audit

We predict the continuous CCME_Values field on [0,100]. CCME_WQI is a redundant class label and never enters a model. The source notebook's objectives are 0.5, 6, 14, 0.05, 8.5, 25, 20 and 45 in the feature order above. These values reproduce the authors' labels and are not asserted to be universal regulatory limits.

Released formula

failed = count(x[j] > objective[j])
excursion = sum(max(x[j] / objective[j] - 1, 0))
F1 = F2 = 100 * failed / 12
nse = excursion / 12
F3 = nse / (0.01 * nse + 0.01)
index = 100 - sqrt($F1^2$ + $F2^2$ + $F3^2$) / 1.732

The reconstruction matches all 2,827,977 source rows within 1e-7, with maximum error 2.84e-14. The denominator of 12 is inferred from exact agreement. The original code uses len(df.columns), making its denominator depend on the frame at execution.

Relationship to official CCME WQI

The official method [2] summarizes scope, frequency and amplitude for defined variables and a sampling period, using suitable upper, lower or range objectives. The source adaptation uses individual rows, F2=F1 and upper-only comparisons, including dissolved oxygen and pH. With eight chemistry variables, a denominator of 12 is not a count of measured variables.

Interpretation

High ML scores reflect approximation of a known calculation from its own components. Temporal separation limits future-data contamination but cannot remove that structural dependence. Computing the transparent source formula is preferable when all eight complete inputs are available.

3. Preprocessing and experimental design

We parse dates strictly as day-month-year, retain original row IDs and remove exact duplicate rows. We never impute targets or invalid dates. All delivered cells are filled, but the source notebook documents upstream imputation. Original missingness and detection-limit flags cannot be reconstructed.

Quality rules mask negative concentrations, nonfinite readings and pH outside [0,14]. Temperature outside [-5,50] Celsius triggers a review flag. This is a declared screening range, not a universal physical limit. We do not guess Fahrenheit conversions. High pollution concentrations remain in the experiment.

Each reading receives a missing/invalid flag. Training medians impute masked inputs. Trees use those imputed readings and flags. MLP and Ridge additionally use asinh followed by training-fitted standard scaling. The target remains the original released index.

Partition	Dates	Available rows	Rows used
Train	1940-2018	2,478,761	300,000
Validation	2019	87,177	50,000 tuning / full selection
Calibration	2020	49,182	All
Test	2021-2023	202,886	All

Every model shares the same seed-42 training sample. Validation chooses candidate checkpoints, model family and blend weight. Calibration only sets residual radii. Final test predictions follow frozen choices. Historical sites may recur: 85.9% of test rows have a Country+Area label seen before 2019.

4. Models and tuning

Mean and Ridge baselines establish reference performance. Ridge uses alpha=10 on the same transformed inputs as the MLP. XGBoost [4] provides a strong threshold-based tabular model. The MLP [5] tests learned nonlinear combinations without imposing a false sequence on feature columns.

XGBoost

Histogram trees, learning rate 0.05, depths 4 or 6, at most 1,000 estimators, minimum child weight 5, subsample 0.9, L2 penalty 5 and early stopping patience 40. The chosen depth-6 model uses 803 boosting rounds.

Neural network

Sixteen inputs comprise eight readings and eight flags. We compare hidden widths 64/32 and 128/64 with ReLU, dropout 0.1 [7] and one linear output. AdamW uses learning rate 0.001 and weight decay 0.0001. We train MSE on targets divided by 100, with batch size 2,048, maximum 60 epochs and validation patience 10. The selected 128/64 model reaches the epoch budget. Outputs are clipped to [0,100].

Family	Candidate	Validation RMSE	Best round/epoch
XGBoost	4	0.4781	999
XGBoost	6	0.4527	803
MLP	[64, 32]	1.1211	59
MLP	[128, 64]	0.9371	60

The blend searches 21 weights on full validation predictions. The selected prediction equals 0.9 times XGBoost plus 0.1 times MLP. Candidate counts are bounded and equal by family, but this is not exhaustive tuning or an equal-compute benchmark.

5. Results and model comparison

MAE and RMSE use index points, where lower is better. RMSE weights larger errors more strongly. R-squared compares squared error with the test-target mean and may be negative. Model selection uses full-validation RMSE, never final test scores.

Model	MAE	RMSE	R-squared
Mean baseline	11.7909	13.4584	-0.41409
Ridge	4.3442	5.5676	0.75799
XGBoost	0.1099	0.4262	0.99858
MLP	0.4466	0.9161	0.99345
Blend	0.1267	0.4222	0.99861

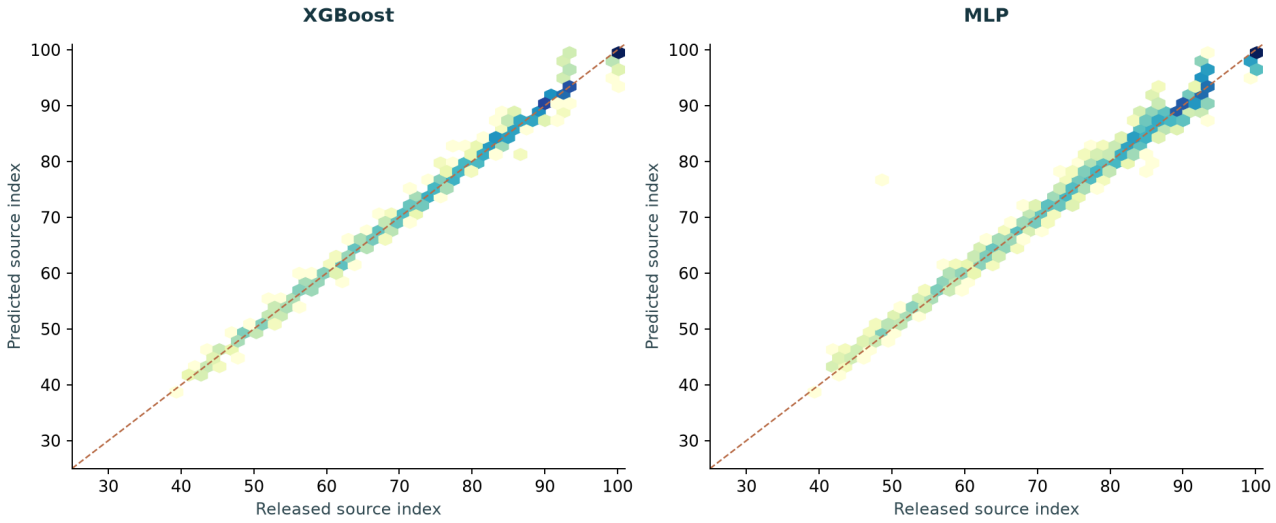


Figure 2. Prediction density for a seeded test sample. Dashed lines denote exact reproduction.

XGBoost wins the individual-model comparison. The blend slightly improves RMSE from 0.4262 to 0.4222, but increases MAE from 0.1099 to 0.1267. The observed change is small and one seed does not support a significance claim. The deterministic formula has test MAE approximately 1.18e-15, reinforcing the emulation interpretation.

6. Feature explanations

Permutation importance shuffles each measurement and its quality flag together on 2,000 validation rows, with three repeats. It measures the increase in validation MAE, allowing a comparison in index units. Native TreeSHAP attributes the selected XGBoost rounds without a separate SHAP package.

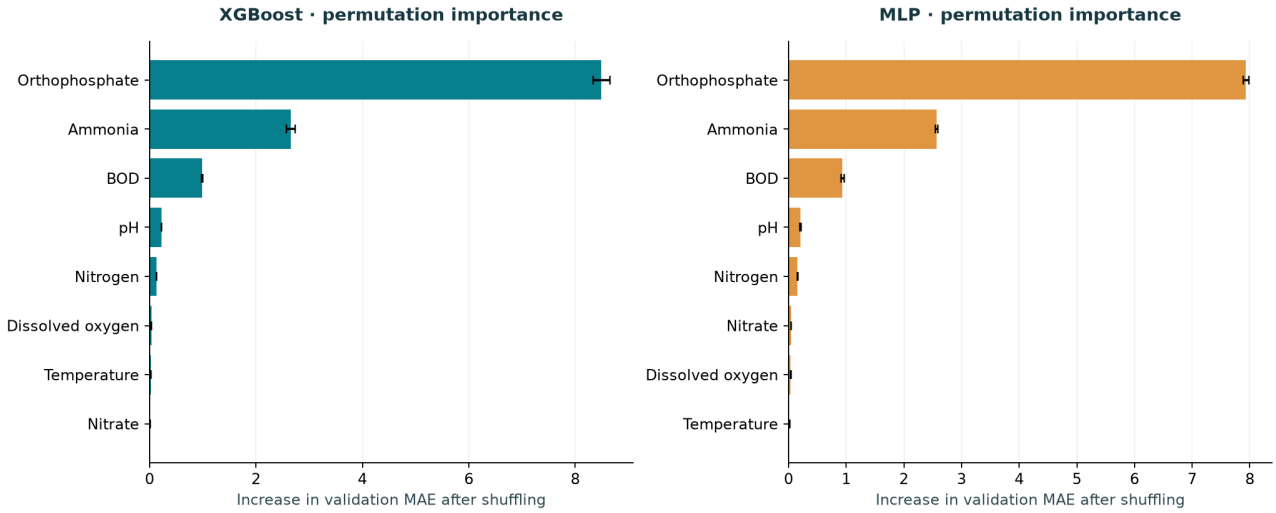


Figure 3. Validation permutation importance. Error bars summarize repeat variation.

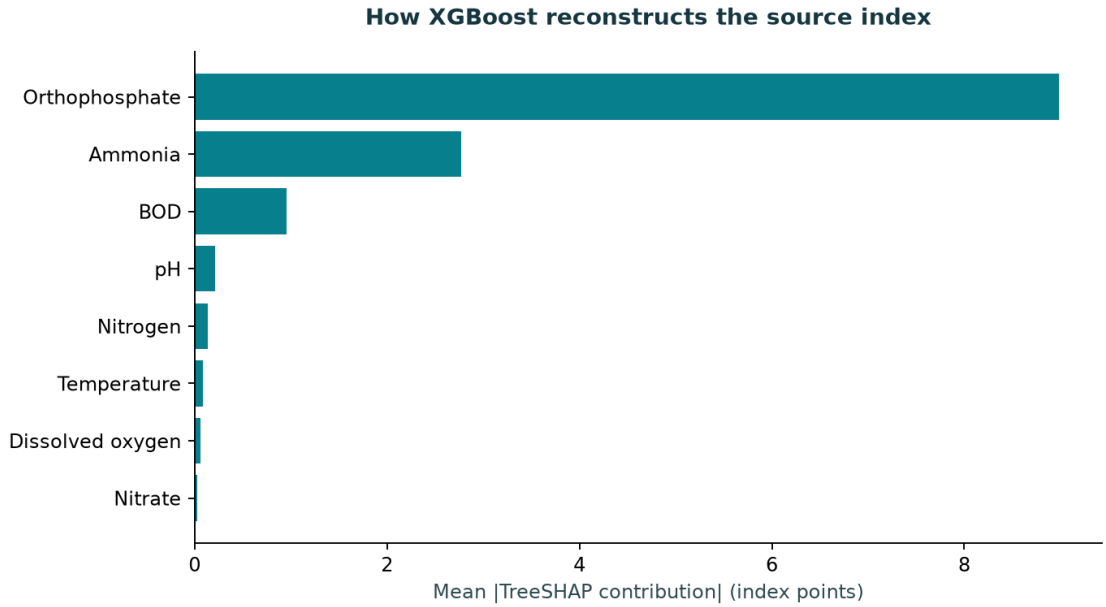


Figure 4. Mean absolute native TreeSHAP contribution. Missing-flag contributions are retained in the CSV.

Orthophosphate and ammonia dominate model reliance. Both directly enter the target formula. Correlation, harmonization and imputation affect attribution. Neither permutation importance nor SHAP establishes that changing a measurement causes a corresponding environmental outcome.

7. Uncertainty and subgroup reliability

Using 2020 held-out data, we take a finite-sample-corrected 90th percentile of absolute residuals [8]. Adding and subtracting this radius forms a constant-radius interval, clipped to [0,100]. Formal split-conformal guarantees depend on exchangeability. Temporal dependence and distribution shift make empirical coverage the relevant evidence here.

Model	Test coverage	Mean width
Mean baseline	94.5%	31.7458
Ridge	84.2%	13.9561
XGBoost	88.4%	0.3073
MLP	90.3%	2.2596
Blend	89.3%	0.4002

Selected blend slice	Rows	MAE	Coverage
Poor	756	0.4934	51.9%
Marginal	8,033	0.4026	45.2%
Fair	18,266	0.2818	66.1%
Good	82,227	0.1606	89.6%
Excellent	93,604	0.0401	97.7%

The blend covers 89.3% overall, but only 45.2% of Marginal rows and 51.9% of Poor rows. A global radius fails to provide uniform conditional reliability. Canada has 46 test observations with 47.8% blend coverage. China has no test rows. New missing-sensor patterns are not covered by this validation.

8. Conclusion, deliverables and references

The project meets the brief as a documented continuous-index comparison: strong AML baseline, regularized neural comparator, chronological split, train-only preprocessing, regression metrics and feature explanations. The target audit is essential to the conclusion. The results support formula emulation, not independent environmental prediction, new-site generalization or water-safety decisions.

The submission uses four folders: Datasets, Notebooks, Reports and figures. Three executed notebooks contain the complete EDA, preprocessing, training, evaluation and prediction workflow. Datasets holds the original CSV archive and one ZIP of trained models and measured outputs. Reports holds this report, the literature review and presentation files. All exported charts are in figures.

Future work should obtain independent outcomes or a properly defined site-window index, preserve original sampling and censoring flags, prespecify a reduced sensor panel and repeat rolling evaluation across seeds. A more elaborate neural model cannot repair a target-definition problem.

References

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[8] Angelopoulos, A. N., and Bates, S. A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. 2021, revised 2022. https://arxiv.org/abs/2107.07511.

The Flight_Delay_Prediction repository supplied by the user informed the submission organization only. No flight data or evaluation results enter this project. https://github.com/abhi-7-7/Flight_Delay_Prediction